

Shiuan-Ting Lin (Jeremy)

Taipei, Taiwan Email LinkedIn GitHub

Profile

Solution Architect at Nokia with a background in mathematics and statistics, currently working on AI-assisted network management, network digital twins, and real-time anomaly detection. Previous academic work includes convolution-based spatial representation analysis, explainable AI for black-box models, and coursework in self-supervised video representation learning. Interested in extending this background toward **self-supervised and predictive representation learning, video understanding, and world models**.

Professional Experience

Nokia May 2026 – Present
Solution Architect, AIOps & Network AI Taipei, Taiwan

- **Agentic AIOps for network management:** Serve as the technical bridge between customers and R&D for AI-assisted network management systems. Define operational use cases, translate customer troubleshooting workflows into product requirements, and validate agentic AI behaviors through end-to-end test scenarios.
- **Network digital twin:** Develop containerlab-based virtual telecom network environments for validating network automation, troubleshooting workflows, and AI-assisted network operations without relying on physical lab devices.
- **Real-time anomaly detection:** Work on anomaly detection and router health monitoring using real-time network telemetry, with the goal of identifying abnormal network behavior and providing interpretable health information to network operators.

Nokia Nov 2024 – Apr 2026
Network Engineer, AI/ML Systems Projects Taipei, Taiwan

- **RAG-based product knowledge system:** Built a retrieval-augmented generation system over 15 long-form telecom product knowledge sources, each approximately 50–100 pages, reducing typical information lookup time from 30–60 minutes to 10–30 seconds and achieving over 90% pass rate on 50–100 validation questions.
- **Capacity intelligence prototype:** Independently developed an AI prototype over 5.93M hourly observations from 16,482 4G cells and 223 KPIs, covering capacity forecasting, congestion-risk identification, anomaly analysis, automated root-cause investigation, and AI-assisted KPI exploration.

Ericsson Jan 2024 – Jun 2024
Master Thesis Student, Trustworthy / Explainable AI Stockholm, Sweden

- Conducted thesis research at Ericsson R&D on generating interpretable rule-based explanations for black-box machine learning models, using telecom HTTP-delay prediction as the application.
- Developed a post-hoc explanation pipeline that generated counterfactual samples around individual predictions, estimated their local probability distributions, and extracted human-readable decision rules using fuzzy labeling and decision trees.
- Evaluated the extracted explanations against existing rule-based explanation methods in terms of how faithfully they reproduced the original model's predictions, how often valid explanations could be generated, and the complexity of the resulting rules.

Education

National Yang Ming Chiao Tung University 2021 – 2024
M.S. in Statistics GPA: 3.96 / 4.30 Taiwan

- Thesis: *Spatial Sparse Principal Component Analysis Using Autoencoders with Convolutions* [Thesis] [Code].
- Studied the effect of introducing convolutional layers into sparse autoencoder-based PCA for spatially structured data, focusing on whether convolution helps preserve local spatial structure in learned sparse representations.
- Conducted 1D linear, 2D linear, and nonlinear simulation experiments and compared convolutional and non-convolutional variants with PCA, Sparse PCA, and sparse autoencoder baselines.
- Applied the framework to monthly sea surface temperature data over the equatorial Pacific Ocean from 2003–2020.

KTH Royal Institute of Technology 2023 – 2024
Exchange Program in Computer Science Stockholm, Sweden

- Degree project conducted at Ericsson R&D: *Explanation Analysis Using Rule Extraction* [Thesis].
- Coursework included deep learning, reinforcement learning, and image analysis and computer vision, with practical work in self-supervised learning, semi-supervised learning, and model interpretability.

National Tsing Hua University

B.S. in Mathematics

2017 – 2021

Taiwan

Selected Academic Projects

Siamese Masked Autoencoder (SiamMAE) Reproduction – KTH Course Project [Code] PyTorch

- Reproduced SiamMAE as the final project for KTH's Advanced Deep Learning course to study self-supervised representation learning from video.
- Used UCF-101 for video pretraining and evaluated learned representations on DAVIS-2017 video object segmentation examples.
- Explored masked prediction across temporally separated video frames as a mechanism for learning visual correspondence without dense supervision.

Deep Learning Coursework – KTH [Code] PyTorch

- Implemented coursework on self-supervised learning with SimCLR, semi-supervised learning with FixMatch, visual explanation methods including Grad-CAM and integrated gradients, and deep-network analysis.

Camera-Based Self-Driving Robot [Code] PyTorch, OpenCV, ROS

- Developed a camera-based lane-following system that processes visual input to predict lane direction and integrates the predictions with ROS-based robot control.

Awards

2nd Place, Nokia MS Espresso AI Competition 2026

Independently developed the Capacity Intelligence prototype; ranked 2nd among 100+ participants.

Gold Award, Intelligent Manufacturing Big Data Analytics Contest – 1st Place Nationwide

Led a two-person team to 1st place among approximately 200 teams with an EfficientNet-based defect classification system.

Honorable Mention, SinoPac AI GO Competition – Top 10 / 800+ Teams

Co-developed a hybrid machine-learning and rule-based rare-event prediction system; achieved a private leaderboard F1 score of 0.8825.

Skills

Programming: Python, C++, R, Git, Linux

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, FLAML, Computer Vision, Self-Supervised Learning, Explainable AI, Forecasting, Anomaly Detection

Statistics: Statistical Modeling, Simulation, Regression, Time Series Analysis, Optimization, PCA, Sparse PCA, Cross-Validation

AI Systems: RAG, Agentic AI, MCP, LangChain, FastAPI, Docker

Network Systems: Containerlab, Network Automation, AIOps, Network Telemetry, Telecom Network Management